

Lesson 5 · Homework

NCEA Level 2 · AS 91262 · Mixed revision and Excellence questions

Time: about 60 minutes

Answer version · tutor's working, not an official mark scheme

Show all working. Justify every maximum or minimum; give units and meaning for rates. [A] Achieved, [M] Merit, [E] Excellence.

1. Differentiate $y = (x + 2)(x^2 - 3)$. [A]

$$y = x^3 + 2x^2 - 3x - 6$$

$$\frac{dy}{dx} = 3x^2 + 4x - 3$$

2. Find the turning points of $y = x^3 - 6x^2 + 9$ and justify their nature. [A]

$$\frac{dy}{dx} = 3x^2 - 12x = 3x(x - 4) = 0, \text{ so } x = 0 \text{ or } x = 4$$

$$y(0) = 9; \quad y(4) = 64 - 96 + 9 = -23$$

$$\frac{d^2y}{dx^2} = 6x - 12: \text{ at } x = 0 \text{ it is } -12 < 0, \text{ so } (0, 9) \text{ is a maximum}$$

$$\text{at } x = 4 \text{ it is } 12 > 0, \text{ so } (4, -23) \text{ is a minimum}$$

3. $f'(x) = 6x - 2$ and the curve passes through $(-1, 8)$. Find $f(x)$. [A]

$$f(x) = 3x^2 - 2x + c; \quad 3 + 2 + c = 8, \text{ so } c = 3$$

$$f(x) = 3x^2 - 2x + 3$$

4. $f'(x) = 4 - x^2$. Without drawing, describe the turning points of f : where they are, and which is a maximum. [A]

$$f'(x) = 0 \text{ when } x = 2 \text{ or } x = -2$$

$$\text{At } x = -2, f' \text{ changes from } - \text{ to } +: \text{ minimum}$$

$$\text{At } x = 2, f' \text{ changes from } + \text{ to } -: \text{ maximum}$$

5. The tangent to $y = x^2 + bx + c$ at the point $(1, 3)$ has gradient 5. Find b and c . [M]

$$\text{Gradient: } \frac{dy}{dx} = 2x + b; \text{ at } x = 1: 2 + b = 5, \text{ so } b = 3$$

$$\text{Point: } 1 + 3 + c = 3, \text{ so } c = -1$$

6. A particle has acceleration $a = 4 - 2t \text{ m s}^{-2}$, starts at the origin, and has velocity 5 m s^{-1} when $t = 0$. Find its maximum velocity and its displacement at that moment. [M]

$$v = 4t - t^2 + 5 \text{ (using } v(0) = 5\text{)}$$

$$\text{Maximum velocity when } a = 0: t = 2 \text{ s; } v(2) = 8 - 4 + 5 = 9 \text{ m s}^{-1} \text{ (maximum: } a \text{ changes from + to -)}$$

$$s = 2t^2 - \frac{1}{3}t^3 + 5t \text{ (from the origin); } s(2) = 8 - \frac{8}{3} + 10 = 15.3 \text{ m}$$

Question 7 [E]

- (iii) The number of daily viewers for an experienced presenter on a video gaming streaming service can be modelled by the following equation:

$$V = 1.6t^3 - 130t^2 + 2900t \quad \{0 \leq t \leq 48\}$$

Where V represents the number of daily viewers and t represents time in months.

Once the stream reaches 10 000 daily viewers, it becomes monetised (the presenter earns money). If the daily viewership falls below 10 000, the presenter will lose this income and will no longer make any money.

Use calculus methods to determine if the stream, after it becomes monetised, ever stops earning money for the presenter.

$$V' = 4.8t^2 - 260t + 2900 = 0$$

$$t = 15.71 \text{ or } t = 38.46 \text{ months (quadratic formula or EQUA)}$$

$$V'' = 9.6t - 260: V''(15.71) = -109 < 0, \text{ local maximum } (V = 19\,678)$$

$$V''(38.46) = 109 > 0, \text{ local minimum}$$

$$\text{Monetised when } V \text{ first reaches } 10\,000: t = 4.20 \text{ months}$$

$$\text{After that the lowest point is the minimum: } V(38.46) = 10\,264$$

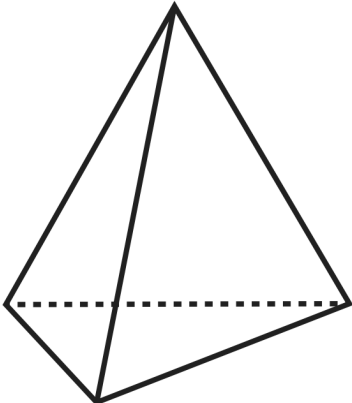
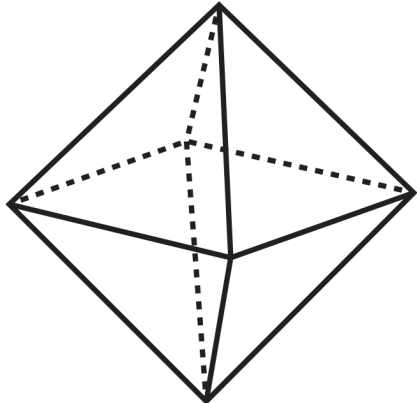
$$\text{End of the domain: } V(48) = 16\,627$$

$$10\,264 > 10\,000, \text{ so daily viewers never fall below } 10\,000 \text{ after monetising:}$$

$$\text{the stream never stops earning money in the first } 48 \text{ months.}$$

Question 8 [E]

- (c) A calculus class is exploring 3-D geometrical art. The students decide that they want to create wireframe models of a tetrahedron and an octahedron, similar to those shown below.

Tetrahedron	Octahedron
	
Surface area = $\sqrt{3}a^2$ where a represents the length of each edge of the tetrahedron	Surface area = $2\sqrt{3}b^2$ where b represents the length of each edge of the octahedron

They will make each edge with a piece of wire, and join each piece together, and then cover each surface with paper to create lanterns similar to those below.



Source: <https://timesofindia.indiatimes.com/city/ahmedabad/20-unique-diy-mathematical-lamps-of-paper-by-ccl-iitgn/>

They wish to create each model so that the total resulting surface area of the two shapes is a minimum.

They have a total of 180 cm of wire to cut and use for both models.

Determine the lengths of the edges, a and b , required to minimise the total surface area of the two shapes when all of the wire is used.

Use calculus to justify that the surface area is a minimum.

Tetrahedron 6 edges, octahedron 12 edges: $6a + 12b = 180$, so $a = 30 - 2b$

$S = \sqrt{3}a^2 + 2\sqrt{3}b^2 = \sqrt{3}(30 - 2b)^2 + 2\sqrt{3}b^2$

(continued on the working page)

Question 8 continued working space

$S'(b) = 2\sqrt{3}(30 - 2b)(-2) + 4\sqrt{3}b = \sqrt{3}(12b - 120)$

$S'(b) = 0$ when $b = 10$, so $a = 30 - 2(10) = 10$

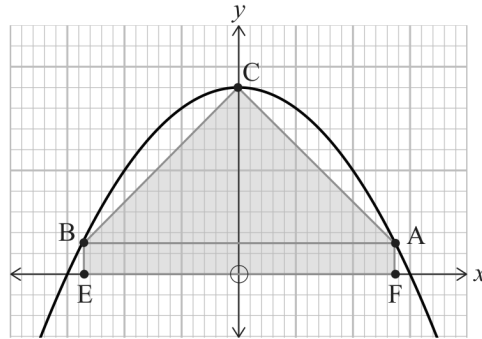
$S''(b) = 12\sqrt{3} > 0$, so the total surface area is a minimum

$a = 10$ cm and $b = 10$ cm (minimum total surface area $300\sqrt{3} = 519.6$ cm²)

Question 9 Extension [E]

(ii) For the generalised problem where the house shape is bounded by a parabola where:

- C is at $(0, d)$.
- A and B are on the parabola $y = d - kx^2$.
- E and F are on the x -axis.



Show that the maximum area enclosed by the house shape occurs when the area of the rectangular section ABEF and the area of the triangular section ABC are equal.

A is $(x, d - kx^2)$, $x > 0$. Wall height $d - kx^2$; triangle height $d - (d - kx^2) = kx^2$

Rectangle = $2x(d - kx^2) = 2dx - 2kx^3$; triangle = $x \cdot kx^2 = kx^3$

Area A = $2dx - 2kx^3 + kx^3 = 2dx - kx^3$

$A' = 2d - 3kx^2 = 0$, so $kx^2 = \frac{2d}{3}$

$A'' = -6kx < 0$ (as $k, x > 0$), so this is a maximum

At the maximum: rectangle = $2x(d - kx^2) = 2x\left(d - \frac{2d}{3}\right) = \frac{2dx}{3}$

Triangle = $x \cdot kx^2 = x \cdot \frac{2d}{3} = \frac{2dx}{3}$

So the rectangle and the triangle have equal areas at the maximum.